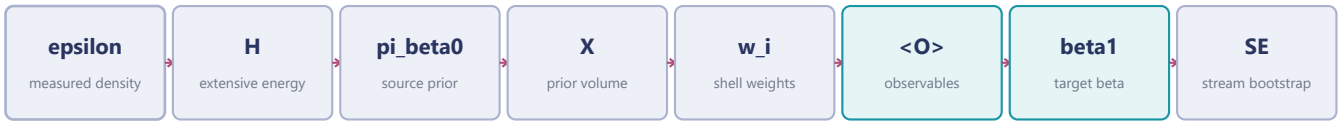
THEORY REFERENCE

Nested sampling theory in equations

A compact derivation of the source-prior transform, deterministic shrinkage, stable observables, beta-only reweighting, and stream-bootstrap errors.

From measured density to uncertainty



Scope

This document follows the implemented convention: the canonical ensemble at beta_0 is the normalized nested-sampling prior. Reweighting changes beta only; theta remains fixed.

Symbol map

Symbol	Meaning	Symbol	Meaning
phi	lattice configuration	epsilon	measured intensive action density
H	beta-independent extensive energy	beta_0, beta_1	source and target couplings
K_eff	combined live-point count	X	source-prior mass below a constraint
w_i	dead-shell quadrature mass	O	generic observable
B	bootstrap replicate count	ESS_quad	quadrature weight concentration

THEORETICAL STEP 1

Define the source measure and constraint coordinate

The simulation starts from a canonical source ensemble, so nested sampling reorganizes an already normalized probability measure.

Action normalization

$$V = L^2, \qquad \varepsilon(\phi) = \frac{S_\beta(\phi)}{2\beta N V}$$

$$H(\phi) = 2NV\varepsilon(\phi), \qquad S_\beta(\phi) = \beta H(\phi)$$

Canonical source prior

$$\pi_{\beta_0}(d\phi) = \frac{e^{-\beta_0 H(\phi)}}{Z(\beta_0)}\,\mu(d\phi), \qquad \int \pi_{\beta_0}(d\phi) = 1$$

Constraint coordinate

$$X(h) = \int \mathbf{1}[H(\phi) < h]\,\pi_{\beta_0}(d\phi)$$

The dead point is the largest-energy live point. As the constraint decreases, the retained region H < h shrinks and X moves from one toward zero.

Expectation as a one-dimensional prior-volume integral

$$\langle O \rangle_{\beta_0} = \int O(\phi)\,\pi_{\beta_0}(d\phi) = \int_0^1 O(X)\,dX$$

Why the source likelihood is 1

The Boltzmann factor at beta_0 already belongs to the normalized prior. In nested-sampling evidence notation, the additional likelihood is L(X)=1, so its integral is one. This does not yield the physical Z(beta_0), which is already hidden in the prior normalization.

$$Z_{\mathrm{NS,\,source}} = \int_0^1 1\,dX = 1 \qquad \Longrightarrow \qquad \log Z_{\mathrm{NS,\,source}} = 0$$

THEORETICAL STEP 2

Replace stochastic shrinkage by deterministic log shrinkage

After merging independent streams by decreasing energy, the effective live-point count fixes the expected compression per dead point.

Combined live-point count

$$K_{\mathrm{eff}} = \sum_{s=1}^{S} K_s$$

Shrinkage distribution

$$t_i = \frac{X_i}{X_{i-1}} \sim \mathrm{Beta}(K_{\mathrm{eff}}, 1), \qquad \mathbb{E}[\log t_i] = -\frac{1}{K_{\mathrm{eff}}}$$

$$\log t = -\frac{1}{K_{\mathrm{eff}}}, \qquad t = \exp\left(-\frac{1}{K_{\mathrm{eff}}}\right)$$

Index convention and dead-shell masses

$$X_{-1} = 1, \qquad X_i = t^{i+1}, \qquad w_i = X_{i-1} - X_i = t^i(1-t)$$

$$\sum_{i=0}^{M-1} w_i = 1 - t^M = 1 - X_{M-1}$$

Log-domain construction

$$\log(1-t) = \log[-\mathrm{expm1}(\log t)]$$

$$\log w_i = i \log t + \log(1-t), \qquad \log M_{\mathrm{dead}} = \mathrm{LSE}_i(\log w_i)$$

$$\log \widetilde{w}_i = \log w_i - \log M_{\mathrm{dead}}$$

Coverage identity

The expected remaining live mass is X_final=t^M. Therefore M_dead + X_final = 1. Report log X_final even when X_final underflows in linear floating point.

THEORETICAL STEP 3

Compute observables and fluctuations

Normalized shell masses define all source-beta estimates. Non-negative quantities admit a fully logarithmic mean; fluctuations use centered observations.

Normalized weighted mean

$$\widetilde{w}_i = \frac{w_i}{\sum_j w_j}, \qquad \langle O \rangle_{\beta_0} \approx \sum_i \widetilde{w}_i O_i$$

Non-negative observable in log space

$$\log \langle O \rangle = \mathrm{LSE}_i(\log \widetilde{w}_i + \log O_i), \qquad O_i \geq 0$$

Signed decomposition

$$A_+ = \mathrm{LSE}_{O_i > 0}(\log \widetilde{w}_i + \log O_i), \qquad A_- = \mathrm{LSE}_{O_i < 0}(\log \widetilde{w}_i + \log |O_i|)$$

$$\langle O \rangle = e^{A_+} - e^{A_-}$$

Centered second moment

$$\mathrm{Var}(O) = \sum_i \widetilde{w}_i (O_i - \langle O \rangle)^2$$

Topological susceptibility

$$\langle Q_U^2 \rangle = \sum_i \widetilde{w}_i Q_{U,i}^2, \qquad \chi_{\mathrm{top}} = \frac{\mathrm{Var}(Q_U)}{L^2}$$

Second-moment correlation length

$$\xi_L^2 = \frac{G(0)/G(k) - 1}{4 \sin^2(\pi/L)}, \qquad \xi_L = \sqrt{\xi_L^2}$$

Numerical implementation

Representable signed products are summed directly to preserve cancellation accuracy. Only products that underflow are reconstructed in signed log space; log_mean remains fully logarithmic.

THEORETICAL STEP 4

Change beta by a density ratio

The beta-independent extensive energy converts the source quadrature at beta_0 into a target quadrature at beta_1. No other parameter is reweighted.

Density ratio

$$\Delta\beta = \beta_1 - \beta_0, \qquad \frac{\pi_{\beta_1}(\phi)}{\pi_{\beta_0}(\phi)} = \frac{Z(\beta_0)}{Z(\beta_1)} e^{-\Delta\beta H(\phi)}$$

Partition-function ratio

$$R_{10} = \frac{Z(\beta_1)}{Z(\beta_0)} = \langle e^{-\Delta\beta H} \rangle_{\beta_0} \approx \sum_i w_i e^{-\Delta\beta H_i}$$

$$\log W_i = \log w_i - \Delta\beta H_i, \qquad \log R_{10} = \mathrm{LSE}_i\,(\log W_i)$$

Target weights and observables

$$\log \widetilde{W}_i = \log W_i - \log R_{10}, \qquad \langle O \rangle_{\beta_1} \approx \sum_i \widetilde{W}_i O_i$$

Quadrature concentration

$$\mathrm{ESS}_{\mathrm{quad}} = \frac{1}{\sum_i \widetilde{W}_i^2}$$

Fixed theta

$$S(\beta, \theta) = \beta H - \theta Q, \qquad S(\beta_1, \theta) - S(\beta_0, \theta) = \Delta\beta H$$

Overlap condition

Reweighting is trustworthy only when the source quadrature resolves the region carrying the target weights. ESS_quad diagnoses concentration but is not an independent-sample count and does not replace bootstrap uncertainty.

Thermodynamic interpretation

$$\frac{d \log Z}{d\beta} = -\langle H \rangle_\beta, \qquad \frac{d^2 \log Z}{d\beta^2} = \mathrm{Var}_\beta\,(H)$$

THEORETICAL STEP 5

Bootstrap independent streams

Each stream is one correlated block. Resampling whole streams preserves within-stream dependence and repeats the full nonlinear merge-and-weight analysis.

Block resampling

$$I_{b1}, \ldots, I_{bS} \overset{\text{iid}}{\sim} \text{Uniform}\{1, \ldots, S\}$$

$$\mathcal{D}^{(b)} = \{\mathcal{D}_{I_{b1}}, \ldots, \mathcal{D}_{I_{bS}}\}, \qquad K_{\text{eff}}^{(b)} = \sum_{i=1}^{S} K_{I_{bj}}$$

Rebuild the estimator

$$\widehat{T}^{(b)} = T\big(\text{sort}_H \mathcal{D}^{(b)},\ w^{(b)},\ \widetilde{W}^{(b)}\big)$$

For every replicate: concatenate the selected streams, sort again by decreasing energy, recompute K_eff, rebuild source weights, optionally reweight beta, and recompute all derived quantities.

Bootstrap standard error

$$\overline{T}^{*} = \frac{1}{B}\sum_{b=1}^{B} \widehat{T}^{(b)}, \qquad \widehat{\text{SE}}(T) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B} (\widehat{T}^{(b)} - \overline{T}^{*})^2}$$

Why dead points are not resampled

Dead points inside a stream share Markov-chain and nested-sampling history. Treating them as independent would destroy this dependence and generally underestimate uncertainty.

Compact equation chain

$$\varepsilon \to H \to \pi_{\beta_0} \to X \to w_i \to \langle O \rangle_{\beta_0} \to \widetilde{W}_i \to \langle O \rangle_{\beta_1} \to \widehat{\text{SE}}$$

Diagnostic	Equation-level interpretation
log_accumulated_prior_mass	log(1 - X_final): source-prior mass represented by dead shells.
log_remaining_prior_mass	log X_final: unresolved expected live mass.
log_Z_ratio	log[Z(beta_1)/Z(beta_0)]: physical beta ratio estimated from source quadrature.
ESS_quad	Inverse squared-weight concentration at target beta.
bootstrap error	Between-stream resampling uncertainty of the full estimator.

Core assumptions

Independent seeds define independent streams; all runs share lattice size, source beta, and theta; N matches the compiled executable; the run is long enough that X_final is negligible for the target quantity; and target-beta overlap is adequate.